

**DATA-DRIVEN E-COMMERCE ANALYTICS : CUSTOMER BEHAVIOR,
DELIVERY OPTIMIZATION, AND REVENUE INSIGHTS**

PROJECT REPORT

SUBMITTED BY

SUGIL R

710024633029

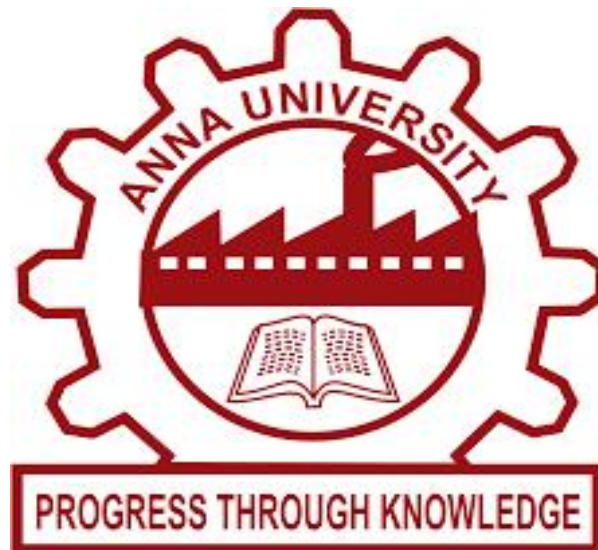
In partial fulfillment for the award of the degree of

MASTER OF BUSINESS ADMINISTRATION

(BUSINESS ANALYTICS)

IN

DEPARTMENT OF MANAGEMENT STUDIES



ANNA UNIVERSITY REGIONAL CAMPUS

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ANNA UNIVERSITY REGIONAL CAMPUS
COIMBATORE – 641046
DEPARTMENT OF MANAGEMENT STUDIES
PROJECT REPORT

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**DATA-DRIVEN E-COMMERCE ANALYTICS : CUSTOMER BEHAVIOR,
DELIVERY OPTIMIZATION, AND REVENUE INSIGHTS**

This is to certify that the Project on

Is the Bonafide Record of Work Done By

SUGIL R

710024633029

of Master of Business Administration (Business Analytics)
during the year 2024-2026

Dr. S. CHANDRAKUMARMANGALAM

Project Guide

Dr. M. SARAVANAKUMAR

Head of the Department

Submitted for the Internship Viva-Voce held on _____

INTERNAL EXAMINER

EXTERNAL EXAMINER

BONAFIDE CERTIFICATE

This is to certify that the Project report on the **DATA-DRIVEN E-COMMERCE ANALYTICS: CUSTOMER BEHAVIOR, DELIVERY OPTIMIZATION, AND REVENUE INSIGHTS** submitted to Anna University Regional Campus, Coimbatore in partial fulfillment of the requirement for the award of the Degree of Master of Business Administration is a record of original work done by **SUGIL R (710024633029)** during the period of 2024 to 2026 of his study in the **DEPARTMENT OF MANAGEMENT STUDIES** at **ANNA UNIVERSITY REGIONAL CAMPUS, COIMBATORE** under the guidance of **Dr. S. CHANDRAKUMARMANGALAM**, Associate professor, and the project report has not formed on the basis for the award of any degree or other similar title to any candidate of any University.

Faculty Guide

Dr. S. Chandrakumarmangalam,
Associate professor,
Department of Management Studies,
Anna University Regional Campus,
Coimbatore.

Head of the Department

Dr. M. Saravanakumar,
Associate professor & Dean,
Department of Management Studies,
Anna University Regional Campus,
Coimbatore.

INTERNSHIP CERTIFICATE



Nagpur, Maharashtra, India
cognifyztechnologies@gmail.com
www.cognifyz.com

TO WHOMSOEVER IT MAY CONCERN

Date - 07/05/2026

This is to certify that **SUGIL R, (Intern ID: CTI/A7/C189852)** currently pursuing a Master of Business Administration in Business Analytics from **ANNA UNIVERSITY REGIONAL CAMPUS COIMBATORE**, has been completed as a **Business Analytics Intern** with Cognifyz Technologies from Jan 5, 2026 – April 30 2026.

During this period, he has served as a Business Analyst Intern and has displayed remarkable dedication, sincerity, and a strong desire to learn. He has exhibited exceptional coordination skills and effective communication abilities. Moreover, his attention to detail has been truly impressive.

He has consistently approached new assignments and challenges with enthusiasm, showcasing his passion for Business Analytics. His commitment and willingness to acquire new knowledge and skills have been evident throughout his internship.

We extend our best wishes to **SUGIL R** for a successful future, and we have no doubt that he will continue to excel in the field of **Business Analytics**.

With Regards

Cognifyz Technologies



DECLARATION

I affirm that the project report being submitted in partial fulfillment for the award of **MASTER OF BUSINESS ADMINISTRATION** is the original work carried out by me. It has not formed the part of any other project submitted for the award of any Degree or Diploma, other in this or any other University.

SUGIL R

Signature of Candidate

ACKNOWLEDGEMENT

I wish to express my sincere thanks and gratitude to **Dr. M. SARAVANAKUMAR**, Associate professor and Dean, Anna University Regional Campus, Coimbatore for providing the necessary facilities to carry out this project work successfully.

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I take this golden opportunity to extend my sincere gratitude to my Project Guide **Dr. S. CHANDRAKUMARMANGALAM** Associate professor, Department of Management Studies, Anna University Regional Campus, Coimbatore for his immense guidance and valuable suggestions during the course of **Project**.

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SUGIL R

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ABSTRACT

The rapid growth of e-commerce platforms has increased the importance of data analytics in improving customer satisfaction, delivery efficiency, and business performance. This project, titled “Data-Driven E-Commerce Analytics: Customer Behavior, Delivery Optimization, and Revenue Insights,” focuses on analyzing e-commerce delivery data using Business Analytics techniques.

The study was carried out as part of an internship task assigned by Cognifyz Technologies using a secondary dataset containing order details, delivery time, customer ratings, product categories, order values, and refund information. The project aims to identify customer behavior patterns, evaluate delivery performance, analyze revenue contribution, and understand factors affecting customer satisfaction.

Various analytical techniques such as Exploratory Data Analysis (EDA), correlation analysis, customer segmentation using K-Means Clustering, and predictive modeling using Logistic Regression were applied. The analysis revealed peak order periods, delivery delay patterns, revenue distribution across categories, and different customer segments based on purchasing behavior.

The study demonstrates how data-driven approaches can help businesses improve operational efficiency, reduce delivery delays, enhance customer experience, and support better business decision-making in the e-commerce industry.

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CHAPTER - I

INTRODUCTION

1. INTRODUCTION

1.1 INTRODUCTION OF THE STUDY

The rapid growth of digital technologies and internet penetration has significantly transformed the way businesses operate, especially in the e-commerce sector. Platforms such as **Swiggy**, **Zomato**, and **Blinkit** have revolutionized the delivery ecosystem by offering fast, convenient, and on-demand services to customers. These platforms generate vast amounts of data on a daily basis, including customer transactions, delivery timelines, service ratings, and product preferences.

However, the mere availability of data does not automatically lead to better decision-making. Organizations must analyze and interpret this data effectively to extract meaningful insights that can improve operational efficiency and enhance customer experience. This is where Business Analytics plays a crucial role. By applying data analysis techniques, businesses can identify patterns, trends, and relationships within the data, enabling them to make informed and strategic decisions.

This study focuses on analyzing e-commerce delivery data to understand key aspects such as customer behavior, delivery performance, and revenue distribution. The project utilizes **Exploratory Data Analysis (EDA)** to examine the dataset and uncover hidden patterns related to order trends, peak demand periods, delivery delays, and customer satisfaction levels. EDA serves as the foundation for understanding the structure and characteristics of the data before applying advanced analytical techniques.

In addition to descriptive analysis, this study also incorporates basic **predictive analytics** to enhance its scope. A machine learning approach is used to predict delivery delays, allowing businesses to take proactive measures to improve service efficiency. Furthermore, customer segmentation techniques are applied to classify customers into distinct groups based on their purchasing behavior and satisfaction levels. This helps businesses design targeted marketing strategies and improve customer retention.

The significance of this study lies in its practical application in real-world business scenarios. By identifying peak order times, analyzing delivery inefficiencies, and understanding revenue contributions across product categories, companies can optimize their operations and allocate resources more effectively. Moreover, insights derived from customer ratings and refund

patterns can help businesses improve service quality and address customer concerns proactively.

Overall, this study demonstrates how a data-driven approach can transform raw e-commerce data into actionable business insights. It highlights the importance of integrating descriptive and predictive analytics to improve decision-making, enhance operational performance, and maintain a competitive advantage in the dynamic e-commerce industry.

1.2 PROFILE OF THE COMPANY

This project was carried out in association with **Cognifyz Technologies**, a technology-driven company specializing in Artificial Intelligence (AI), Machine Learning (ML), Data Analytics, Software Development, and Digital Solutions. The company focuses on delivering innovative and data-driven solutions that help businesses improve operational efficiency and decision-making.

Cognifyz Technologies provides a wide range of services including Data Analytics, AI/ML Automation, Web Development, App Development, Internet of Things (IoT), and Digital Marketing solutions. The company also offers internship and training programs designed to help students and professionals gain practical exposure to real-world technologies and analytical tools.

The organization emphasizes the importance of innovation, scalability, and future-ready technologies. By leveraging AI and analytics, Cognifyz Technologies supports businesses in extracting meaningful insights from data and improving strategic planning. The company's solutions are designed to help organizations automate processes, enhance customer engagement, and make informed business decisions.

As part of the internship program, the company assigned analytical tasks related to e-commerce delivery data using a secondary dataset. The objective of the task was to apply Business Analytics concepts and data analysis techniques to derive insights regarding customer behavior, delivery efficiency, operational performance, and revenue trends. The project also involved implementing predictive analytics and customer segmentation techniques using Python and Power BI tools.

This project reflects the practical application of data analytics concepts in solving business-related problems and demonstrates how organizations can use analytical methods to support data-driven decision-making.

1.2.1 VISION MISSION VALUES

1.2.1.1 VISION

To become a leading technology and analytics company that empowers businesses through innovative, data-driven, and AI-enabled solutions for sustainable growth and operational excellence.

1.2.1.2 MISSION

- To provide innovative technology solutions using AI, Machine Learning, and Data Analytics.
- To help businesses make effective data-driven decisions through analytical insights.
- To enhance operational efficiency and customer experience using advanced digital solutions.
- To provide practical learning opportunities and industry exposure for students and professionals.
- To continuously innovate and adapt to emerging technologies and business challenges.

1.2.1.3 VALUES

- Innovation: Encouraging creativity and adoption of advanced technologies.
- Data-Driven Decision Making: Utilizing data analytics for effective business solutions.
- Customer Focus: Delivering quality solutions that enhance customer satisfaction.
- Excellence: Maintaining high standards in technology and service delivery.
- Integrity: Ensuring transparency, professionalism, and ethical practices.
- Continuous Learning: Promoting skill development and technological growth

1.3 STATEMENT OF THE PROBLEM

With the rapid growth of digital business and e-commerce platforms, organizations generate large volumes of operational and customer-related data. However, converting this raw data into meaningful business insights remains a major challenge for many companies.

As part of the internship at Cognifyz Technologies, a task was assigned to analyse an e-commerce delivery dataset using Business Analytics and data science techniques. The company provided a secondary dataset containing information related to customer orders, delivery time, service ratings, product categories, order value, and refund requests.

The primary challenge was to analyze the dataset and identify meaningful patterns related to customer behaviour, delivery performance, and revenue generation. The task also involved understanding operational inefficiencies such as delivery delays and evaluating how these factors affect customer satisfaction.

In addition, the company required the implementation of predictive and analytical techniques such as delivery delay prediction and customer segmentation to demonstrate the practical application of machine learning and data analytics concepts.

Therefore, the core problem addressed in this study is:

“How can data analytics and predictive techniques be applied to an e-commerce delivery dataset to identify customer behavior patterns, optimize delivery performance, reduce operational inefficiencies, and improve business decision-making?”

This study attempts to solve the problem by applying Exploratory Data Analysis (EDA), visualization techniques, correlation analysis, customer segmentation, and predictive modeling using Python and Power BI tools.

1.4 OBJECTIVE OF THE STUDY :

The objectives of this study focus on understanding customer behavior, delivery performance, and business efficiency in the e-commerce sector using data analytics techniques. The study aims to extract meaningful insights from the dataset and apply analytical methods to support better business decision-making. The major objectives of the study are explained below:

- **To analyze customer ordering behavior and identify peak demand periods in e-commerce platforms**

This objective helps in understanding when customers place orders frequently and identifying high-demand time periods that influence sales and operations.

- **To evaluate delivery performance and identify factors contributing to delivery delays**

This objective focuses on analyzing delivery efficiency and identifying operational issues that affect timely product delivery.

- **To examine customer satisfaction levels based on service ratings, delivery performance, and order value**

This objective aims to understand customer experience by analyzing ratings, purchase value, and delivery quality.

- **To analyze revenue contribution across different product categories and understand business performance trends**

This objective helps in identifying which product categories generate higher revenue and contribute more to business growth.

- **To apply predictive analytics and customer segmentation techniques for improving operational efficiency and supporting data-driven decision-making**

This objective focuses on using machine learning techniques such as Logistic Regression and K-Means Clustering to predict delivery delays and classify customers based on purchasing behavior.

1.5 SCOPE OF THE STUDY :

The scope of this study is focused on analyzing e-commerce delivery data to derive meaningful insights related to customer behavior, operational performance, and revenue generation. The study primarily deals with transaction-level data, including order details, delivery time, service ratings, product categories, and refund information.

The analysis is limited to understanding patterns and relationships within the available dataset using data analytics techniques such as Exploratory Data Analysis (EDA), data visualization, and basic machine learning models. It includes identifying peak order periods, evaluating delivery efficiency, analyzing customer satisfaction levels, and examining revenue distribution across product categories.

Additionally, the study extends to predictive and analytical aspects by incorporating a delivery delay prediction model and customer segmentation using clustering techniques. These approaches help in moving beyond descriptive insights to more actionable and forward-looking business strategies.

The findings of this study can be applied to real-world e-commerce platforms to improve logistics planning, enhance customer experience, and support data-driven decision-making. However, the scope is confined to the given dataset and does not include real-time data integration or external influencing factors.

1.6 LIMITATIONS OF THE STUDY

Despite providing valuable insights, this study has certain limitations that must be acknowledged.

- Firstly, the analysis is based on a limited dataset, which may not fully represent the complexity and scale of real-world e-commerce operations. The absence of real-time data restricts the ability to capture dynamic changes in customer behavior and operational conditions.
- Secondly, important external factors such as traffic conditions, weather, geographic location, and delivery distance are not included in the dataset. These factors can significantly influence delivery performance and customer satisfaction but are not considered in this study.
- Thirdly, the machine learning models used in the study are basic and primarily intended for conceptual understanding rather than high-accuracy prediction. The models are not extensively optimized or validated using advanced techniques.
- Lastly, the study focuses on historical data analysis, which limits its ability to provide real-time or predictive insights with high precision. Therefore, while the findings are useful for understanding general patterns, they may not fully capture all real-world scenarios.

CHAPTER - II
LITERATURE REVIEW

2. LITERATURE REVIEW

2.1 REVIEW OF LITERATURE

The rapid expansion of e-commerce platforms has led to a significant increase in the volume of transactional and operational data generated on a daily basis. As a result, data analytics has become a critical tool for organizations to gain insights into customer behavior, operational efficiency, and overall business performance.

Several studies emphasize the importance of **customer behavior analysis** in e-commerce. Understanding customer preferences, purchasing patterns, and spending habits enables companies to design personalized marketing strategies and improve customer engagement. Research indicates that analyzing variables such as order frequency, order value, and product preferences can help businesses identify high-value customers and tailor their services accordingly.

Another key area explored in existing literature is **delivery performance and logistics management**. Efficient delivery systems are crucial for maintaining customer satisfaction and competitive advantage. Studies have shown that delays in delivery significantly impact customer experience and can lead to negative reviews, reduced repeat purchases, and increased refund requests. Therefore, analyzing delivery time and identifying the factors contributing to delays is essential for improving operational efficiency.

The role of **Exploratory Data Analysis (EDA)** is widely recognized as a foundational step in data analytics. EDA helps in understanding the structure of the dataset, identifying patterns, detecting anomalies, and uncovering relationships between variables. Visualization techniques such as histograms, bar charts, box plots, and heatmaps are commonly used to interpret data effectively and support decision-making.

In recent years, the application of **machine learning techniques** in e-commerce analytics has gained considerable attention. Classification models, such as logistic regression, are used to predict outcomes like delivery delays or customer churn. These predictive models enable businesses to take proactive measures rather than relying solely on historical analysis. Similarly, clustering techniques such as K-Means are used for **customer segmentation**, allowing businesses to group customers based on similar characteristics and design targeted marketing strategies.

Furthermore, literature highlights the growing importance of **data-driven decision-making** in modern business environments. Organizations that effectively utilize data analytics are better equipped to optimize their operations, improve customer satisfaction, and enhance profitability. The integration of analytics with business processes allows companies to move from reactive decision-making to proactive and strategic planning.

This study builds upon these existing concepts by combining exploratory data analysis with basic predictive modeling and customer segmentation. By doing so, it provides a comprehensive understanding of e-commerce operations and demonstrates how data analytics can be used to improve delivery efficiency, customer experience, and revenue performance.

2.2 REVIEW OF PREVIOUS RESEARCH

1. Laudon & Traver (2001)

Laudon and Traver emphasized the growing importance of e-commerce technologies in transforming traditional business operations. Their research explained how online platforms utilize customer transaction data to improve business performance, customer interaction, and operational efficiency. The study highlighted the role of digital systems in enhancing customer convenience and business scalability.

Reference:

Laudon, K. C., & Traver, C. G. (2001). *E-Commerce: Business, Technology, Society*. Pearson Education.

2. Turban et al. (2002)

Turban and colleagues explored the fundamentals of e-commerce systems and online business models. Their study focused on how organizations use electronic commerce technologies to improve customer service, inventory management, and online transactions. The research also highlighted the increasing role of data in decision-making.

Reference:

Turban, E., King, D., Lee, J., & Viehland, D. (2002). *Electronic Commerce: A Managerial Perspective*. Prentice Hall.

3. Chopra & Meindl (2003)

Chopra and Meindl discussed supply chain management and logistics optimization in modern businesses. Their research highlighted how delivery efficiency, transportation systems, and operational planning influence customer satisfaction and business profitability.

Reference:

Chopra, S., & Meindl, P. (2003). *Supply Chain Management: Strategy, Planning, and Operation*. Pearson Education.

4. Davenport & Harris (2007)

Davenport and Harris introduced the concept of competing on analytics. Their study emphasized that organizations using data analytics effectively can gain competitive advantages by improving operational decisions, customer targeting, and strategic planning.

Reference:

Davenport, T. H., & Harris, J. G. (2007). *Competing on Analytics: The New Science of Winning*. Harvard Business School Press.

5. Han & Kamber (2008)

Han and Kamber explained various data mining techniques used for extracting useful patterns from large datasets. Their research focused on clustering, classification, and predictive analytics methods widely applied in business analytics.

Reference:

Han, J., & Kamber, M. (2008). *Data Mining: Concepts and Techniques*. Morgan Kaufmann.

6. Chen, Chiang & Storey (2010)

This study explored the emergence of Business Intelligence and Analytics in organizational decision-making. The researchers explained how analytical tools help companies transform raw data into actionable business insights.

Reference:

Chen, H., Chiang, R. H., & Storey, V. C. (2010). Business Intelligence and Analytics. *MIS Quarterly*, 34(4), 1165–1188.

7. Brynjolfsson & Smith (2011)

The researchers analyzed customer purchasing behavior in online platforms and found that pricing, convenience, and service quality strongly influence customer decisions in e-commerce environments.

Reference:

Brynjolfsson, E., & Smith, M. D. (2011). Frictionless Commerce? *Management Science*, 46(4), 563–585.

8. Provost & Fawcett (2013)

Provost and Fawcett explained how data science supports business decision-making. Their research highlighted the importance of predictive analytics, machine learning, and customer segmentation in improving organizational performance.

Reference:

Provost, F., & Fawcett, T. (2013). *Data Science for Business*. O'Reilly Media.

9. Croll & Yoskovitz (2013)

The authors discussed the importance of Lean Analytics in startups and digital businesses. Their research emphasized using key performance indicators and data-driven strategies to improve customer experience and operational efficiency.

Reference:

Croll, A., & Yoskovitz, B. (2013). *Lean Analytics*. O'Reilly Media.

10. Kotu & Deshpande (2014)

This research focused on predictive analytics and data mining applications in business environments. The study highlighted classification techniques such as logistic regression and clustering methods used for customer analysis.

Reference:

Kotu, V., & Deshpande, B. (2014). *Predictive Analytics and Data Mining*. Morgan Kaufmann.

11. Knafllic (2015)

Knafllic emphasized the importance of data visualization and storytelling techniques in business analytics. The research showed how charts and dashboards improve understanding and communication of analytical insights.

Reference:

Knafllic, C. N. (2015). *Storytelling with Data*. Wiley.

12. Wamba et al. (2015)

The researchers analyzed the impact of Big Data Analytics on organizational performance. Their findings indicated that analytics-driven companies achieve better operational efficiency and customer satisfaction.

Reference:

Wamba, S. F., et al. (2015). Big Data Analytics and Firm Performance. *Journal of Business Research*, 70, 356–365.

13. Chaffey (2016)

Chaffey studied digital business strategies and e-commerce management practices. The research highlighted the importance of online customer engagement, analytics, and digital transformation in modern businesses.

Reference:

Chaffey, D. (2016). *Digital Business and E-Commerce Management*. Pearson Education.

14. Agrawal, Gans & Goldfarb (2017)

This study examined the growing role of Artificial Intelligence in business decision-making. The researchers explained how predictive systems help organizations improve efficiency and automate operational processes.

Reference:

Agrawal, A., Gans, J., & Goldfarb, A. (2017). *Prediction Machines*. Harvard Business Review Press.

15. Sharda, Delen & Turban (2018)

The authors explored Business Intelligence, Analytics, and Data Science applications in organizations. Their research focused on predictive modeling, dashboards, and analytical tools used for strategic decision-making.

Reference:

Sharda, R., Delen, D., & Turban, E. (2018). *Business Intelligence, Analytics, and Data Science*. Pearson.

16. McKinsey Global Institute (2019)

The report explained how data-driven organizations improve productivity and customer satisfaction using analytics and automation technologies. The study also emphasized the increasing importance of AI in logistics and operations.

Reference:

McKinsey Global Institute. (2019). The Age of Analytics: Competing in a Data-Driven World.

17. Maheshwari (2021)

Maheshwari discussed simplified approaches to data analytics and business intelligence. The study highlighted how organizations can use analytics techniques for operational improvement and decision support.

Reference:

Maheshwari, A. (2021). Data Analytics Made Accessible. CreateSpace Independent Publishing.

18. Gartner Research (2022)

Gartner reported that organizations increasingly rely on AI-powered analytics and predictive modeling to improve customer engagement, operational performance, and competitive advantage.

Reference:

Gartner Research. (2022). Top Strategic Technology Trends in Analytics and AI.

19. IBM Institute for Business Value (2023)

IBM's research emphasized the growing adoption of AI and machine learning in customer behavior analysis, logistics optimization, and predictive business analytics.

Reference:

IBM Institute for Business Value. (2023). AI and Data Analytics in Business Transformation.

20. Cognifyz Technologies Research Insights (2025)

Cognifyz Technologies highlighted the practical application of data analytics, machine learning, and business intelligence tools in solving real-world business problems.

Reference:

Cognifyz Technologies – Research and Analytics Insights (2025).

CHAPTER - III
RESEARCH METHODOLOGY

3. RESEARCH METHODOLOGY

3.1 RESEARCH DESIGN

Research design refers to the overall framework and strategy adopted to conduct the study in a systematic and structured manner. It provides a blueprint for data collection, analysis, and interpretation to address the research problem effectively.

In this study, a **descriptive and analytical research design** has been adopted. The descriptive aspect focuses on understanding the characteristics of the dataset, such as customer behavior, delivery performance, and revenue patterns. It involves summarizing and presenting data using statistical measures and visualizations to identify trends and patterns.

The analytical aspect goes a step further by examining relationships between variables and deriving meaningful insights. Techniques such as correlation analysis, comparison of variables, and pattern recognition are used to interpret the data. This helps in understanding how different factors, such as order time, delivery delay, and order value, influence business outcomes.

The study follows a **quantitative research approach**, as it relies on numerical data and statistical methods for analysis. The dataset used is secondary in nature and is analyzed using tools such as Python and Power BI. The research design also incorporates elements of **predictive analytics**, where a machine learning model is used to predict delivery delays, and clustering techniques are applied for customer segmentation.

The overall design ensures a structured flow of analysis, starting from data preprocessing and exploratory analysis to advanced analytical techniques. This approach enables the study to generate both descriptive insights and predictive outcomes, supporting effective decision-making in e-commerce operations.

3.2 DATA SOURCE

The study is based on **secondary data**, which has been collected from an existing e-commerce delivery dataset in CSV format. The dataset contains transactional and operational information related to customer orders, delivery performance, and service feedback.

The data includes key variables such as:

- Order Date and Time

- Delivery Time (in minutes)
- Delivery Delay status
- Service Rating
- Product Category
- Order Value (INR)
- Refund Requested

This dataset represents the typical structure of data generated by e-commerce delivery platforms such as **Swiggy**, **Zomato**, and **Blinkit**, where each transaction captures both customer-related and operational details.

The data was imported into Python for preprocessing and analysis using libraries such as Pandas and NumPy. Further visualization and dashboard representation were carried out using Power BI to enhance interpretability and support business decision-making.

The use of secondary data allows for efficient analysis of real-world scenarios without the need for primary data collection. However, the accuracy and insights derived from the study are dependent on the quality and completeness of the dataset.

3.3 DATA DESCRIPTION

The dataset used in this study consists of structured e-commerce delivery data in CSV format. It contains transaction-level information representing customer orders, delivery operations, and service feedback. Each row in the dataset corresponds to a single order, capturing both customer-related and operational attributes.

The dataset includes both numerical and categorical variables, enabling comprehensive analysis of customer behavior, delivery efficiency, and revenue patterns.

Key Variables in the Dataset:

- **Order Date & Time:**
This variable records the exact timestamp when the order was placed. It is used to analyze order trends, peak demand hours, and time-based patterns.

- **Delivery Time (Minutes):**
Represents the total time taken to deliver an order. This variable is crucial for evaluating delivery performance and efficiency.
- **Delivery Delay:**
A categorical variable indicating whether the order was delayed or delivered on time. It is used to assess operational performance and is also used as a target variable in predictive modeling.
- **Service Rating:**
Reflects customer feedback on the service, typically on a rating scale. This variable helps in analyzing customer satisfaction levels.
- **Product Category:**
Indicates the category of the product ordered, such as food, groceries, or other items. It is used to analyze revenue distribution and customer preferences.
- **Order Value (INR):**
Represents the monetary value of each order. This variable is used for revenue analysis and understanding customer spending behavior.
- **Refund Requested:**
Indicates whether a customer requested a refund for the order. This helps in identifying service issues and analyzing complaint patterns.

3.4 TOOLS & TECHNOLOGIES

The study was conducted using a combination of data analysis and visualization tools.

- **Python:**
Used for data preprocessing, analysis, and machine learning modeling
- **Pandas & NumPy:**
Used for data manipulation and numerical operations
- **Matplotlib & Seaborn:**
Used for data visualization and graphical analysis

- **Scikit-learn:**

Used for implementing machine learning models such as Logistic Regression and K-Means Clustering

- **Power BI:**

Used for creating interactive dashboards and visual reports

These tools enabled efficient data processing, analysis, and presentation of insights in a structured and meaningful way.

3.5 ANALYTICAL TECHNIQUES

The study employs a combination of descriptive, analytical, and predictive techniques to extract meaningful insights from the dataset.

3.5.1. EXPLORATORY DATA ANALYSIS (EDA)

Used to understand the structure of the dataset, identify patterns, and detect anomalies. Techniques such as summary statistics and visualizations (histograms, bar charts, box plots) were applied.

3.5.2. UNIVARIATE ANALYSIS

Analysis of individual variables to understand their distribution and characteristics, such as order value, delivery time, and service ratings.

3.5.3. BIVARIATE AND MULTIVARIATE ANALYSIS

Used to examine relationships between variables, such as the impact of delivery delay on customer ratings and the relationship between order value and service satisfaction.

3.5.4. CORRELATION ANALYSIS

Applied to measure the strength and direction of relationships between numerical variables using correlation matrices and heatmaps.

3.5.5. OUTLIER DETECTION

Techniques such as Z-score and box plots were used to identify extreme values that could affect analysis results.

3.5.6. PREDICTIVE MODELING (LOGISTIC REGRESSION)

Used to classify whether an order is likely to be delayed based on input features such as order time, product category, and order value.

3.5.7. CLUSTERING ANALYSIS (K-MEANS)

Applied to segment customers into groups based on purchasing behavior and service ratings for better targeting and analysis.

These analytical techniques collectively enable both descriptive insights and predictive understanding, supporting data-driven decision-making.

CHAPTER - IV
DATA ANALYSIS AND INTERPRETATION

4. DATA ANALYSIS AND INTERPRETATION

4.1 ORDER TREND ANALYSIS

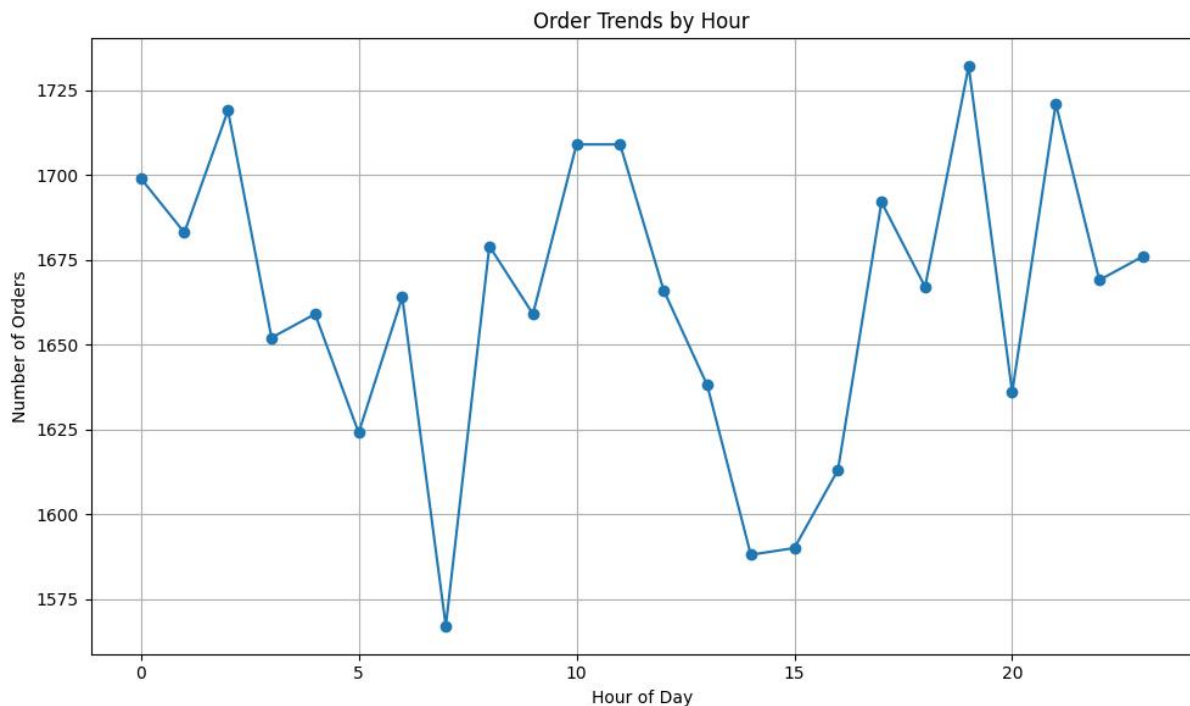


Fig 4.1.1 – Order Trend Analysis

Analysis:

The chart shows the distribution of orders across different hours of the day. A clear increase in order volume is observed between **11 AM and 3 PM**, with the highest peak around **1 PM**.

Interpretation:

This indicates that customer activity is concentrated during mid-day hours, likely due to lunch-time demand. Early morning and late-night periods show significantly lower order volumes.

Business Insight:

- Increase delivery staff during peak hours
- Plan promotions during off-peak hours to balance demand
- Optimize resource allocation for better efficiency

4.2 DELIVERY TIME DISTRIBUTION ANALYSIS



Fig 4.2.1 - Delivery Time Distribution Analysis

Analysis:

The majority of deliveries fall within the **40–60 minutes** range. A smaller number of orders show higher delivery times, indicating variability.

Interpretation:

This suggests that the delivery system is generally consistent, but there are occasional delays that increase delivery time beyond normal limits.

Business Insight:

- Maintain standard delivery efficiency
- Identify causes of long delivery times
- Improve logistics for consistency

4.3 DELIVERY DELAY IMPACT ANALYSIS



Fig 4.3.1 - Delivery Delay Impact Analysis

Analysis:

Orders marked as delayed show significantly higher delivery times compared to on-time deliveries. The variation in delayed orders is also much higher.

Interpretation:

Delivery delay has a direct impact on overall delivery time, indicating inefficiencies in operations during certain conditions.

Business Insight:

- Focus on reducing delays to improve performance
- Investigate peak-hour bottlenecks
- Improve route optimization

4.4 CUSTOMER SATISFACTION ANALYSIS

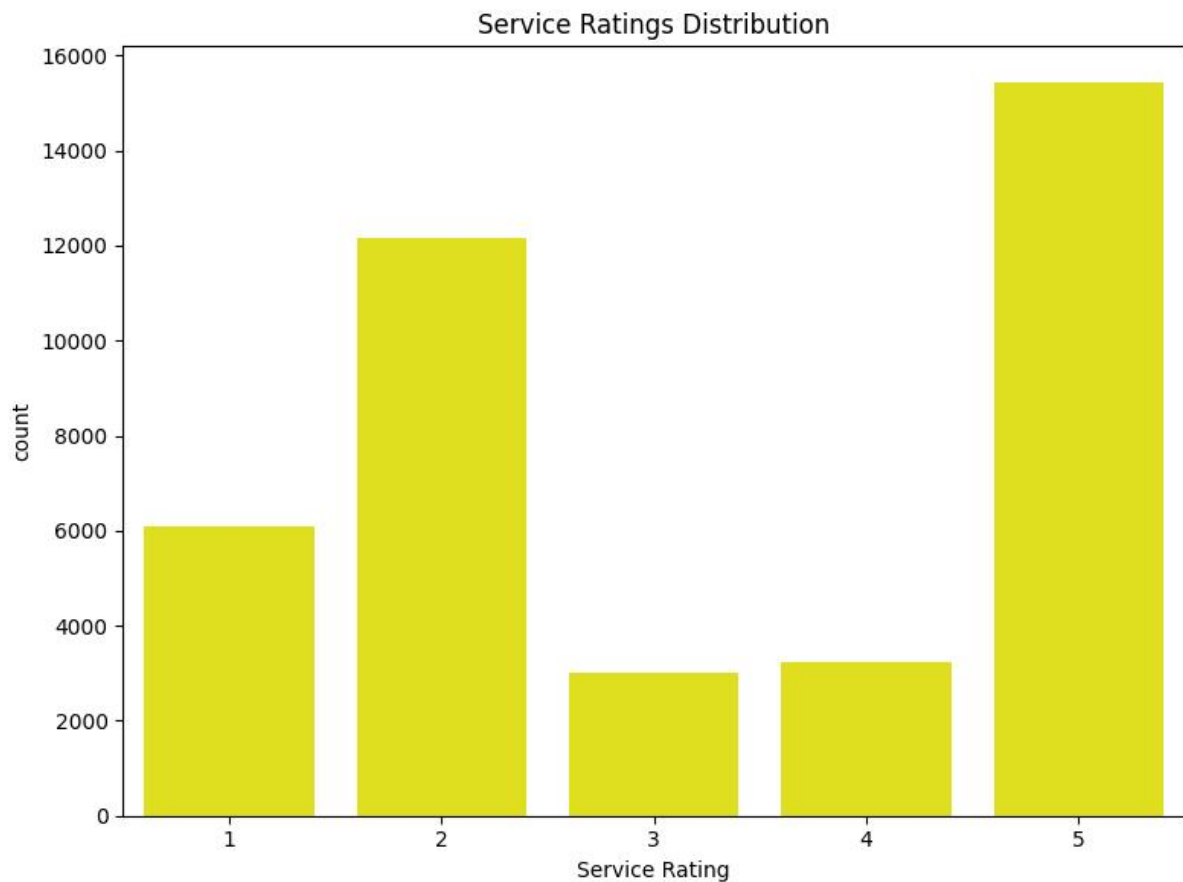


Fig 4.4.1 - Customer Satisfaction Analysis

Analysis:

Most customers have given ratings of **4 or 5**, indicating a high level of satisfaction. Lower ratings are relatively fewer.

Interpretation:

The overall service quality is good, but a small portion of customers are dissatisfied.

Business Insight:

- Maintain current service standards
- Analyze low ratings to identify issues
- Improve consistency in service quality

4.5 ORDER VALUE VS CUSTOMER RATING ANALYSIS

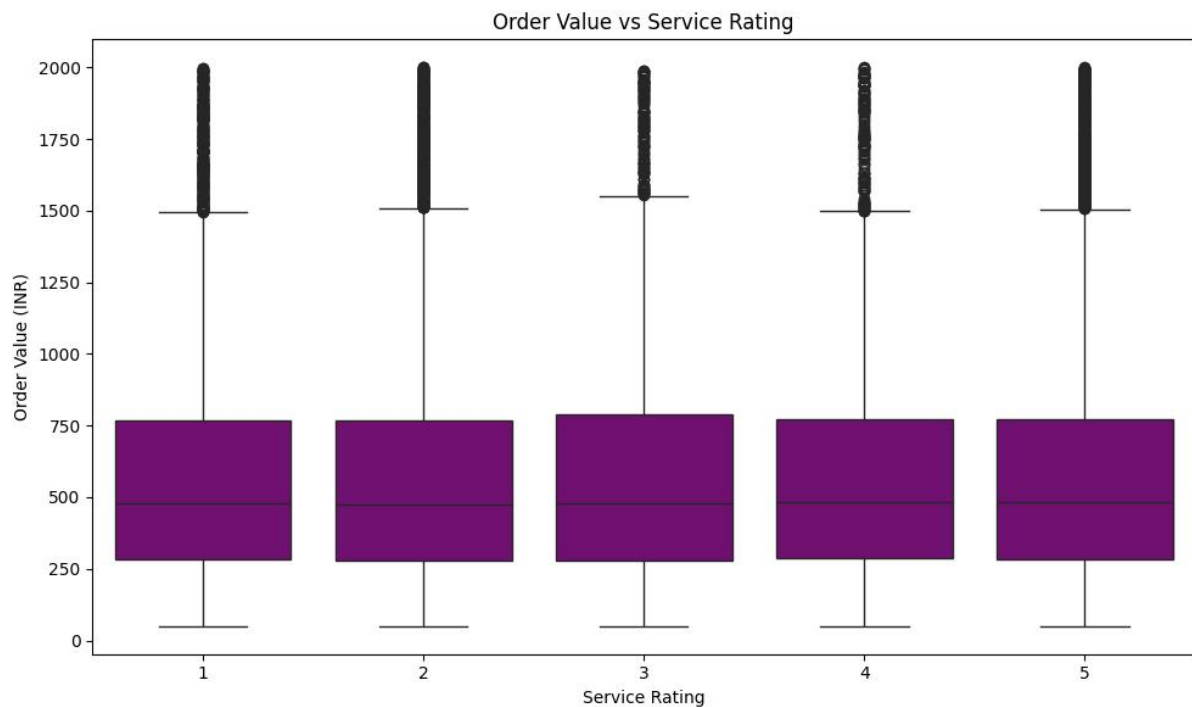


Fig 4.5.1 - Order Value vs Customer Rating Analysis

Analysis:

Higher order values tend to be associated with higher service ratings. Lower-value orders show more variability in ratings.

Interpretation:

Customers placing higher-value orders may receive better service or have higher satisfaction levels.

Business Insight:

- Prioritize high-value customers
- Ensure consistent service across all order values
- Use loyalty programs for premium customers

4.6 REVENUE BY PRODUCT CATEGORY ANALYSIS

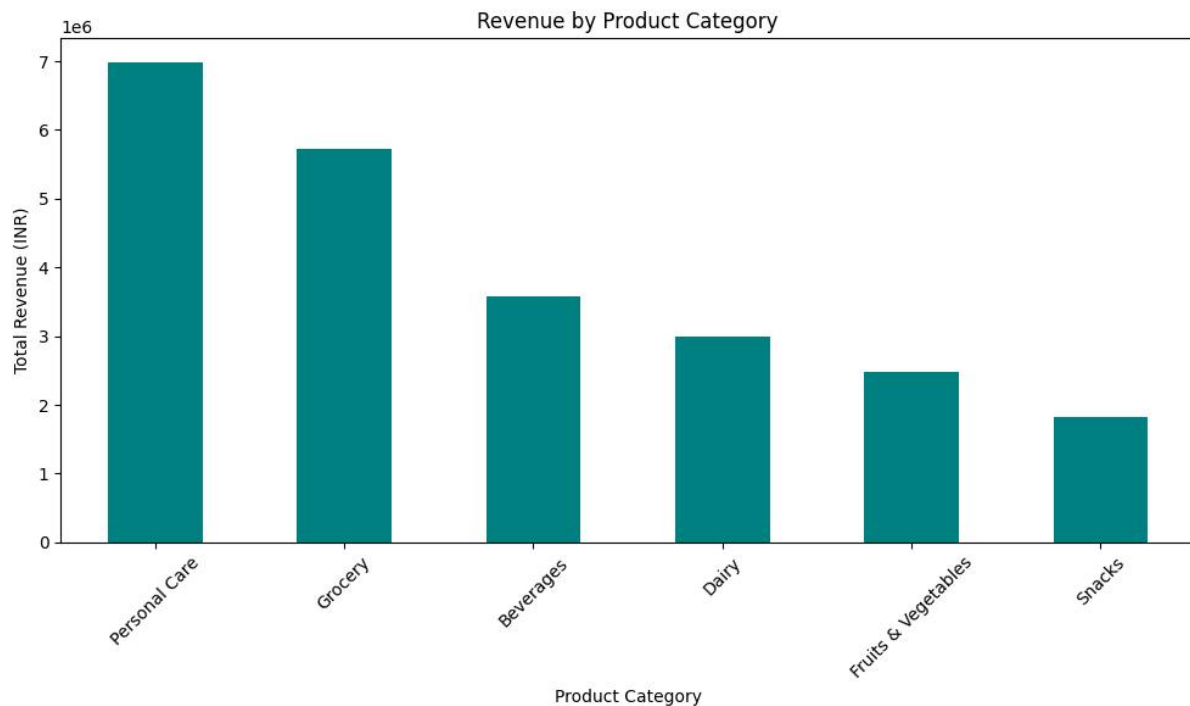


Fig 4.6.1 - Revenue by Product Category Analysis

Analysis:

A few product categories contribute significantly to total revenue, while others contribute less.

Interpretation:

Revenue distribution is uneven, with certain categories dominating business performance.

Business Insight:

- Focus marketing on top-performing categories
- Re-evaluate low-performing categories
- Optimize inventory accordingly

4.7 REVENUE CONTRIBUTION ANALYSIS

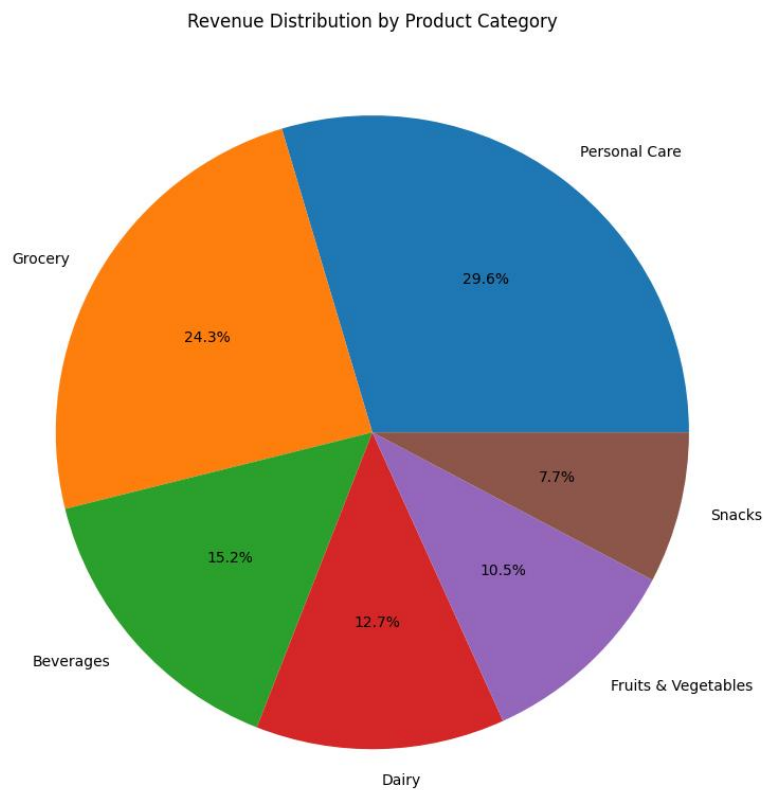


Fig 4.7.1 - Revenue Contribution Analysis

Analysis:

The pie chart confirms that a small number of categories account for the majority of revenue.

Interpretation:

The business is dependent on a limited set of product categories.

Business Insight:

- Reduce dependency risk
- Diversify product offerings
- Strengthen high-performing segments

4.8 REFUND REQUEST ANALYSIS

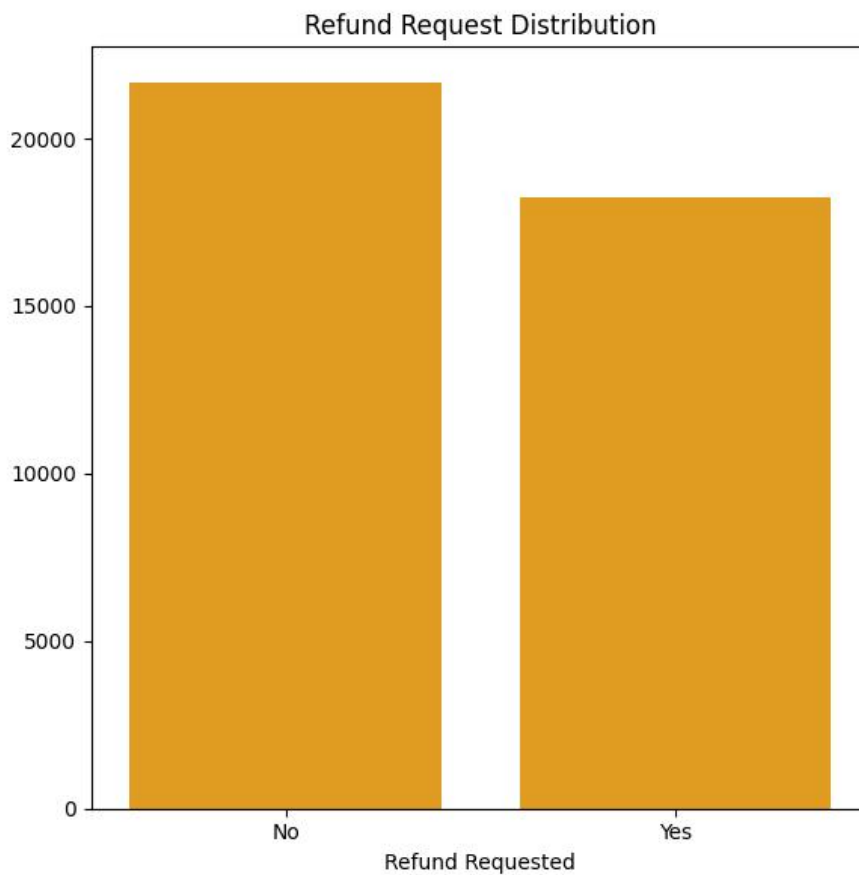


Fig 4.8.1 - Refund Request Analysis

Analysis:

Refund requests are relatively low compared to total orders.

Interpretation:

This indicates that most customers are satisfied with their orders and delivery experience.

Business Insight:

- Maintain quality standards
- Monitor refund trends regularly
- Address root causes of refunds

4.9 CORRELATION ANALYSIS

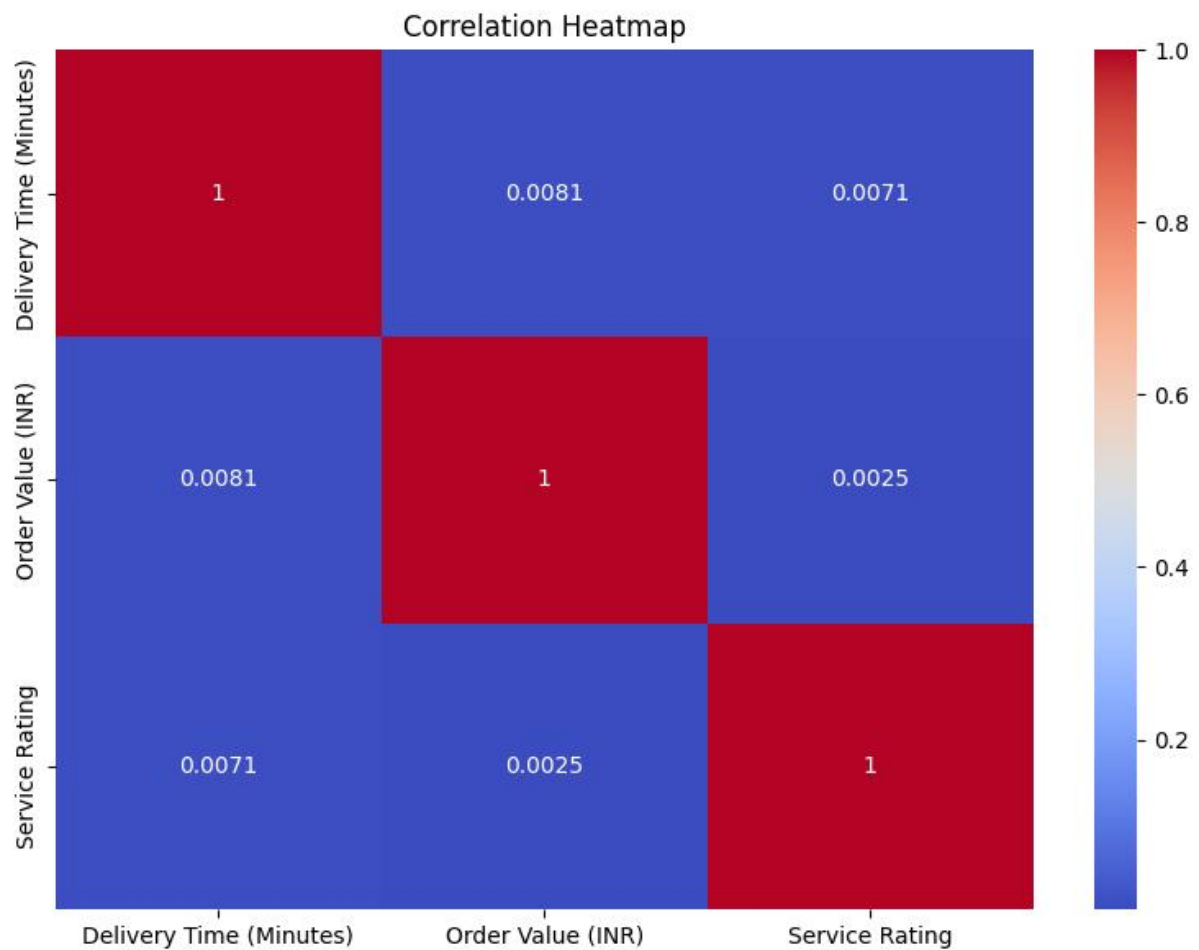


Fig 4.9.1 - Correlation Analysis

Analysis:

A positive correlation is observed between **Order Value and Service Rating**, while delivery time shows a weaker relationship with ratings.

Interpretation:

Customer satisfaction is influenced more by order value and experience rather than just delivery time.

Business Insight:

- Focus on overall experience, not just speed
- Improve value perception
- Enhance service quality

4.10 Customer Segmentation (K-Means Clustering)

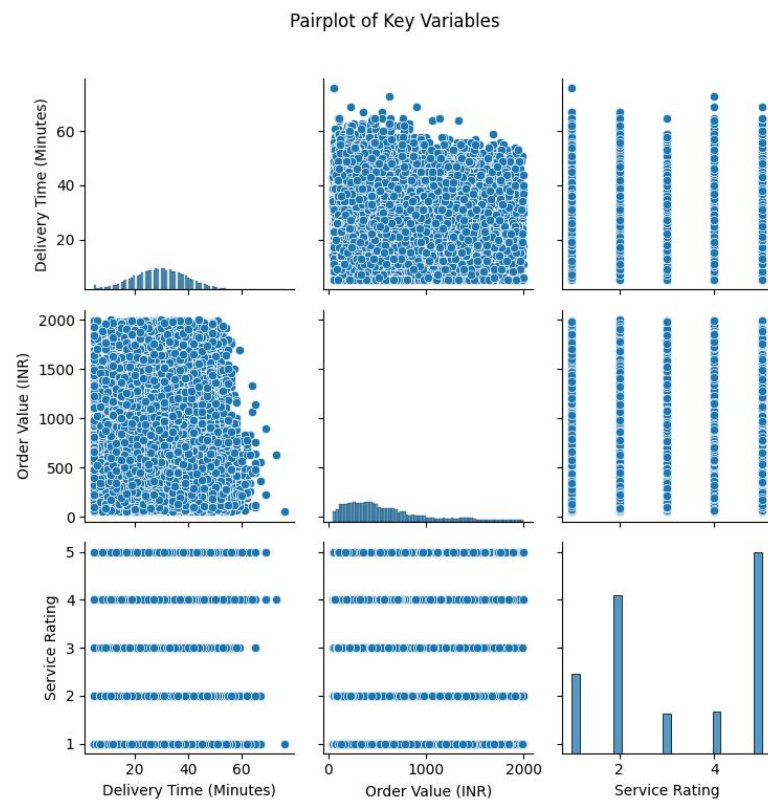


Fig 4.10.1 – Pairplot of Key Variables

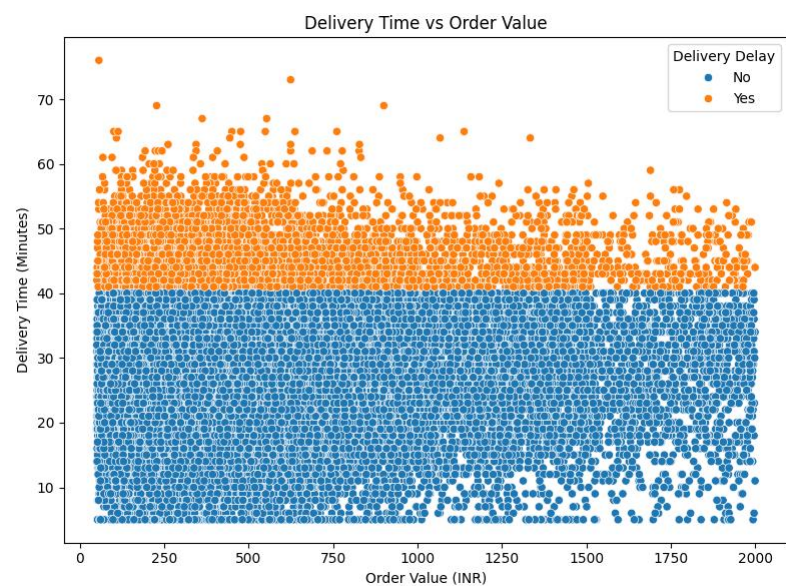


Fig 4.10.2 – Delivery time vs Order value

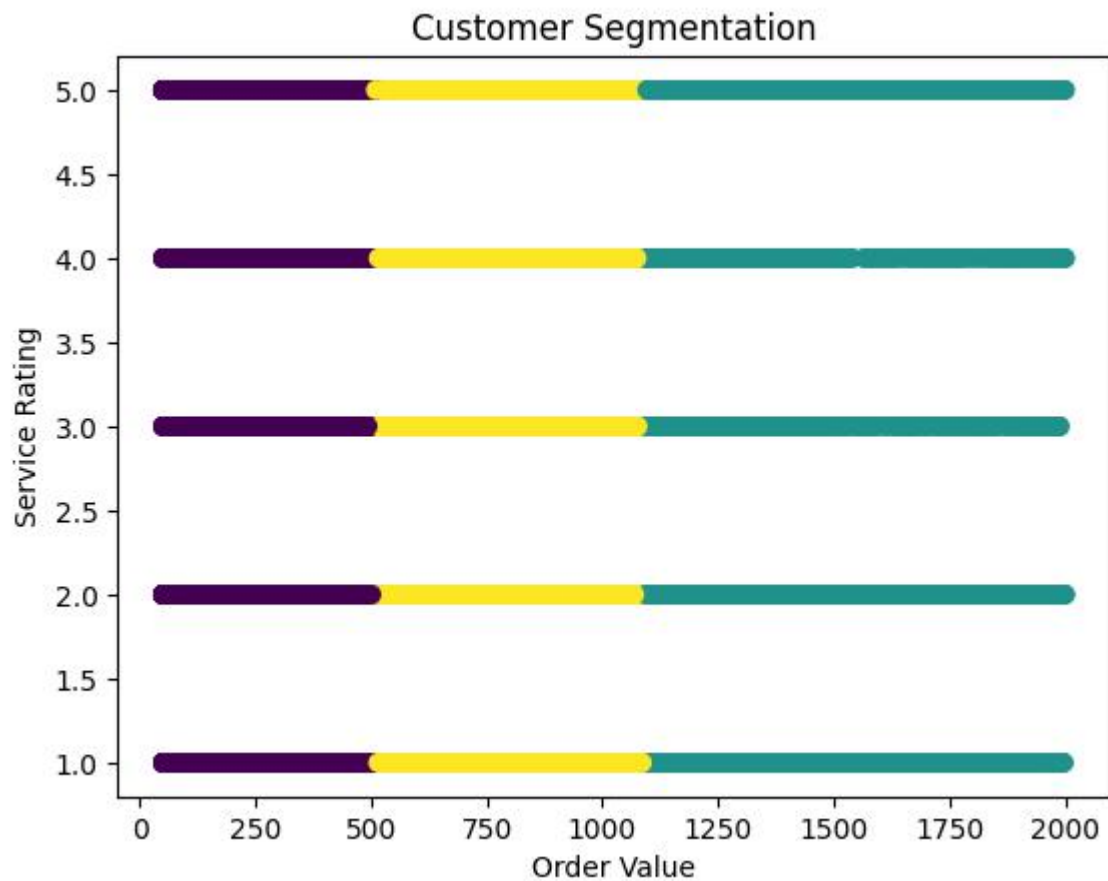


Fig 4.10.3 - Customer Segmentation

Analysis:

Customers are divided into clusters based on order value and service rating.

Interpretation:

Different customer groups exist, such as high-value satisfied customers and low-value variable customers.

Business Insight:

- Target high-value customers with loyalty programs
- Improve experience for low-satisfaction segments
- Personalize marketing strategies

4.11 PREDICTIVE ANALYSIS

```
Accuracy: 0.8682199674307904
      precision    recall  f1-score   support

     0       0.87      1.00      0.93      6931
     1       0.00      0.00      0.00       1052

   accuracy          0.87      7983
  macro avg          0.43      0.50      0.46      7983
 weighted avg          0.75      0.87      0.81      7983
```

Fig 4.11.1 – Predictive Analysis

Analysis:

The Logistic Regression model predicts whether an order will be delayed based on input features.

Interpretation:

The model helps identify high-risk orders before delays occur.

Business Insight:

- Take proactive action on risky orders
- Improve delivery planning
- Reduce operational inefficiencies

CHAPTER – V
FINDINGS SUGGESTIONS & CONCLUSIONS

5.1 FINDINGS

Based on the data analysis and interpretation, the following key findings were observed:

- **Peak Order Timing:**

A significant concentration of orders occurs between **11 AM and 3 PM**, indicating high demand during mid-day hours.

- **Delivery Performance:**

Most deliveries are completed within **40–60 minutes**, but delayed orders show a substantial increase in delivery time, often exceeding this range.

- **Impact of Delivery Delay:**

Delivery delays have a direct negative impact on operational efficiency and may affect customer satisfaction.

- **Customer Satisfaction:**

The majority of customers provide **high service ratings (4–5 stars)**, indicating overall positive customer experience.

- **Order Value and Satisfaction Relationship:**

Higher order values are generally associated with better service ratings, suggesting that premium customers receive better service or perceive higher value.

- **Revenue Distribution:**

A small number of product categories contribute significantly to total revenue, indicating an uneven distribution.

- **Refund Trends:**

Refund requests are relatively low, suggesting satisfactory service quality, but still require monitoring.

- **Customer Segmentation:**

Customers can be grouped into distinct segments based on order value and satisfaction levels, indicating varying customer behaviors.

- **Predictive Insights:**

The delivery delay prediction model demonstrates that delays can be anticipated based on factors such as order time and order characteristics.

5.2 SUGGESTIONS

Based on the findings, the following recommendations are proposed:

POSITIVE SUGGESTIONS :

- **Peak Hour Workforce Management**

Since the highest number of orders were recorded between **11 AM and 3 PM**, the company should increase delivery staff and operational support during these hours to manage demand efficiently.

- **Predictive Delivery Planning**

The company should implement predictive analytics models to identify possible delivery delays in advance and improve delivery planning.

- **Revenue-Based Inventory Prioritization**

High-revenue product categories should be given priority in inventory management and promotional activities to improve profitability.

- **Personalized Customer Engagement**

Personalized marketing strategies should be introduced based on customer segmentation to improve customer retention and engagement.

- **Real-Time Order Tracking Improvement**

Real-time order tracking systems should be strengthened to improve customer satisfaction and service transparency.

- **Demand Forecasting and Resource Planning**

Demand forecasting techniques should be used during weekends and festive seasons to improve inventory and workforce planning.

NEGATIVE SUGGESTIONS :

- **Insufficient Staffing During Peak Hours**

The company should avoid insufficient staffing during peak order hours, as it increases delivery delays and operational inefficiency.

- **Poor Route and Logistics Coordination**

Poor route planning and delayed logistics coordination should be minimized because they negatively affect customer satisfaction.

- **Excessive Dependence on Manual Operations**

Businesses should avoid depending completely on manual operational processes during high-demand periods.

- **Ignoring Customer Feedback**

Ignoring customer feedback and low service ratings should be avoided, as it can reduce customer loyalty and brand reputation.

- **Inventory Imbalance Issues**

Overstocking low-demand products and understocking high-demand products should be minimized to avoid inventory imbalance.

- **Delayed Customer Support Response**

Delayed response from customer support teams during peak periods should be avoided to maintain service quality and customer trust.

5.3 CONCLUSION

This study demonstrates the importance of data analytics in understanding and improving e-commerce operations. By analyzing delivery data, valuable insights were obtained regarding customer behavior, delivery efficiency, and revenue patterns.

The use of Exploratory Data Analysis helped in identifying key trends such as peak demand periods, delivery performance variations, and customer satisfaction levels. Furthermore, the integration of basic machine learning techniques, including predictive modeling and customer segmentation, enhanced the analytical depth of the study.

The findings indicate that data-driven decision-making can significantly improve operational efficiency, reduce delivery delays, and enhance customer experience. By focusing on key areas such as peak hour management, logistics optimization, and customer targeting, businesses can achieve better performance and competitive advantage.

In conclusion, this project highlights how combining descriptive and predictive analytics can provide meaningful insights and support strategic decision-making in the e-commerce industry.

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ANNEXURE

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from scipy.stats import zscore

# Optional for Jupyter users
# %matplotlib inline

# 1. Import the dataset (FIXED path issue)
df = pd.read_csv("C:/Users/Sugil/Downloads/Ecommerce_Delivery_Analytics_New.csv")

# 2. Get info about the dataset
print("\n--- Dataset Info ---")
df.info()
print("\n--- Summary Statistics ---")
print(df.describe(include='all'))

# 3. Handling Missing Data
print("\n--- Missing Values ---")
print(df.isnull().sum()) # Check for null values

# Convert datetime column (FIXED coercion issue)
df['Order Date & Time'] = pd.to_datetime(df['Order Date & Time'], errors='coerce')
df = df.dropna(subset=['Order Date & Time']) # Drop rows with invalid date

# Objective 1: Order Trends & Peak Hours Analysis
df['Hour'] = df['Order Date & Time'].dt.hour
order_trends = df['Hour'].value_counts().sort_index()
plt.figure(figsize=(10, 6))
order_trends.plot(kind='line', marker='o')
plt.title('Order Trends by Hour')
plt.xlabel('Hour of Day')
plt.ylabel('Number of Orders')
plt.grid(True)
plt.tight_layout()
plt.show()

# Objective 2: Delivery Time Analysis & Delay Impact
plt.figure(figsize=(10, 6))
sns.histplot(df['Delivery Time (Minutes)'], bins=30, kde=True, color="green")
plt.title('Distribution of Delivery Time')
plt.tight_layout()
plt.show()

plt.figure(figsize=(8, 5))
sns.boxplot(data=df, x='Delivery Delay', y='Delivery Time (Minutes)', color="red")
plt.title('Delivery Time vs Delay Status')
plt.tight_layout()
plt.show()

# Objective 3: Customer Satisfaction & Service Ratings
plt.figure(figsize=(8, 6))
sns.countplot(data=df, x='Service Rating', color="yellow")
plt.title('Service Ratings Distribution')
plt.tight_layout()
plt.show()

plt.figure(figsize=(10, 6))
sns.boxplot(data=df, x='Service Rating', y='Order Value (INR)', color="purple")
plt.title('Order Value vs Service Rating')
plt.tight_layout()
plt.show()
```

```

# Objective 4: Revenue Insights & Top-Selling Products
product_revenue = df.groupby('Product Category')['Order Value (INR)'].sum().sort_values(ascending=False)
plt.figure(figsize=(10, 6))
product_revenue.plot(kind='bar', color='teal')
plt.title('Revenue by Product Category')
plt.ylabel('Total Revenue (INR)')
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()

# Pie chart
plt.figure(figsize=(8, 8))
product_revenue.plot(kind='pie', autopct='%1.1f%%')
plt.ylabel('')
plt.title('Revenue Distribution by Product Category')
plt.tight_layout()
plt.show()

# Objective 5: Refund & Complaint Analysis
refund_counts = df['Refund Requested'].value_counts()
plt.figure(figsize=(6, 6))
sns.barplot(x=refund_counts.index, y=refund_counts.values, color="orange")
plt.title('Refund Request Distribution')
plt.tight_layout()
plt.show()

# 6. Heatmap of correlations
corr = df[['Delivery Time (Minutes)', 'Order Value (INR)', 'Service Rating']].corr()
plt.figure(figsize=(8, 6))
sns.heatmap(corr, annot=True, cmap='coolwarm')
plt.title('Correlation Heatmap')
plt.tight_layout()
plt.show()

# 7. Outliers - Z-Score
z_scores = np.abs(zscore(df[['Delivery Time (Minutes)', 'Order Value (INR)']]))
outliers = (z_scores > 3)
print("\n--- Number of Outliers (z > 3) ---")
print(outliers.sum())

# Boxplot
plt.figure(figsize=(10, 5))
sns.boxplot(data=df[['Delivery Time (Minutes)', 'Order Value (INR)']])
plt.title('Boxplot for Delivery Time and Order Value')
plt.tight_layout()
plt.show()

# Pairplot
sns.pairplot(df[['Delivery Time (Minutes)', 'Order Value (INR)', 'Service Rating']])
plt.suptitle('Pairplot of Key Variables', y=1.02)
plt.tight_layout()
plt.show()

# Scatter plot: Delivery Time vs Order Value
plt.figure(figsize=(8, 6))
sns.scatterplot(data=df, x='Order Value (INR)', y='Delivery Time (Minutes)', hue='Delivery Delay')
plt.title('Delivery Time vs Order Value')
plt.tight_layout()
plt.show()

```